Koushik Padmanabhan

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EDUCATION

University of Southern California | Viterbi School of Engineering **Aug 2026 – May 2028**
M.S. Computer Science (Game Development) *Los Angeles, CA*
Sri Ramachandra Institute of Higher Education and Research (SRIHER) **Aug 2021 – May 2025**
B.Tech. Computer Science (AI & ML) | CGPA 9.09/10, First Class with Distinction Chennai, India

TECHNICAL SKILLS

Languages: Python, C#, JavaScript, SQL, Perl
Cloud & DevOps: Microsoft Azure (Azure Virtual Machines, Azure App Service, Azure DevOps), AWS (EC2, S3, DynamoDB), Docker, Podman, Git, Grafana, Kibana
AI & ML: LLMs, RAG, AI Agents, Deep Learning, Computer Vision, NLP, Reinforcement Learning, Anomaly Detection
ML Frameworks: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, XGBoost, OpenCV, Ollama
AI-Driven Development: GitHub Spec Kit, Claude Code, GitHub Copilot, OpenAI Codex, Gemini CLI, Google Antigravity
AI Agents & Automation: n8n, Make.com, Retell AI, Power Automate
Web, APIs & Databases: FastAPI, Flask, React, REST APIs, Role-Based Access Control, SharePoint, MSSQL, MySQL
CS Fundamentals: Data Structures, Algorithms, Object-Oriented Design, System Design, Agile/Scrum

EXPERIENCE

Mindera — Software Engineer | *Bangalore, India* **Feb 2025 – Jun 2026**
- Owned **production support** for an enterprise client's **AWS** platform (Amazon EC2, Amazon S3, Amazon DynamoDB), triaging **3 issues weekly** through **Grafana** and **Kibana** logs and **Perl** scripts, shipping fixes within Scrum sprints.
- Led design and delivery of an **LLM-powered** onboarding assistant built on **GitHub's Spec Kit** that answers codebase questions with citations and scaffolds first changes, cutting ramp-up time **50%**.
- Built an **XGBoost** and **scikit-learn** anomaly-detection model and its **data pipeline** for a production fraud service, screening **1,000 transactions daily** at **97% precision**.
- Prototyped emerging **generative AI** capabilities and presented them biweekly to 50+ engineers as working demos.
- Rebuilt the team's **Docker**/**Podman** dev environment after a failed migration, scripting SQL setup to cut setup time by **75%**.

Katpro Technologies — Software Developer (Part-time) | *Bangalore, India* **Dec 2023 – Dec 2024**
- Delivered an **OCR + LLM** pipeline (Python, **MSSQL**) on **Azure Virtual Machines and Azure App Service** that extracts purchase orders, invoices, and GRNs into **SharePoint** via **Power Automate** at **95% field-level accuracy**, saving **20 hours** weekly.
- Architected an end-to-end leasing platform where **AI voice (Retell AI), email, and SMS agents** take renters from first call to signed lease, orchestrated by **Dockerized FastAPI REST APIs** and **Make.com** on Azure.
- Built a **clinical coding** assistant (**React**/JavaScript, **FastAPI**, Ollama) on an Azure Virtual Machine that uses OCR and a locally hosted LLM with **RAG** to assign ICD-10 and CPT codes.
- Set up **CI/CD** in **Azure DevOps** and refactored the codebase into modular **Git** repos and **object-oriented** components.
- Automated candidate screening with an **n8n** + **LLM** workflow that scores and routes applicants, cutting time-to-shortlist by **60%**.
- Resolved a **critical client production issue** within **1 day**, earning an **HR commendation**.

PROJECTS & RESEARCH

FocusNet: Deep Learning Framework for Fundus Pathology | *TensorFlow, CNNs* **Springer, CVR 2025**
- **First-author paper, published by Springer:** designed an attention-augmented **DenseNet169** (**TensorFlow**) classifying fundus images into **4 classes** at **95% accuracy**, best of **12** benchmarked configurations; F1 **0.79 → 0.92**.

Reinforcement Learning & Game AI Agents | *PyTorch, Stable-Baselines3, Unity ML-Agents, C#* **koushik2k3.github.io**
- Taught agents to play games from scratch, progressing from tabular **Q-learning** (FrozenLake, Taxi) to a **DQN** that plays Atari Space Invaders from raw pixels and **PPO** for LunarLander; published 7 agents with evaluation scores on Hugging Face.
- Engineered **Game AI** in **Unity ML-Agents**: a dog that learns to fetch, a target shooter, and a curiosity-driven puzzle explorer.

COMMUNITY SERVICE

- **Mindera: Conducted robotics and coding workshops for middle school students**, **volunteered with NGOs** for neurodivergent and differently abled people, **helped sponsor newborn eye surgeries**, and **represented the company** in Chennai's largest walkathon for the physically challenged.
- **SRIHER:** Wrote and performed theatre for **pediatric patients** at Sri Ramachandra Hospital as **Lead Scriptwriter** of theatre club, organized seminars and talks as **board member** IoT Club and **organized blood drives**.
- **Qwings IIT Madras: Built ML Algorithms as easy to understand websites** and presented them at **high school AI workshops.**

AWARDS, ACTIVITIES & CERTIFICATIONS

- **Certifications: AWS Academy Machine Learning Foundations**; **Hugging Face Deep Reinforcement Learning.**
- **1st Prize**, University **Hackathon**: Meme Explainer Bot, OCR + NLP (OpenCV, Hugging Face Transformers).
- **1st Prizes, Long Jump** for **4 continuous years** in High School.
- **Member**, USC **Shinkendo** Club.